

Quantum Field Theory – Homework 8

Name: _____ Due Date: 2025.12.22

These problems are based on Lectures 15, 16, 17 and 18, focusing on Generating functionals, Wicks theorem, Feynman diagrams, symmetry factors, momentum-space rules, and LSZ reduction. Show key steps clearly and state any assumptions you use.

Problem 1. Generating Functional and Wicks Theorem in Field Theory (15 points)

Consider a free real scalar field $\phi(x)$ in four-dimensional spacetime with action

$$S_0[\phi] = \int d^4x \left(-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right),$$

and define the generating functional

$$Z[J] = \int \mathcal{D}\phi \exp \left(i S_0[\phi] + i \int d^4x J(x) \phi(x) \right).$$

(a) (5 points) Show explicitly that the generating functional can be written as

$$\frac{Z[J]}{Z_0} = \exp \left(\frac{i}{2} \int d^4x d^4y J(x) K^{-1}(x, y) J(y) \right),$$

where

$$K(x, y) = (-\partial^2 + m^2 - i\epsilon) \delta^{(4)}(x - y),$$

and $Z_0 = Z[J = 0]$.

(b) (5 points) Using functional derivatives with respect to $J(x)$, compute the two-point function

$$G_2(x_1, x_2) = \langle 0 | T \phi(x_1) \phi(x_2) | 0 \rangle,$$

and show that

$$G_2(x_1, x_2) = G_F(x_1 - x_2) = -i K^{-1}(x_1, x_2),$$

where G_F is the Feynman propagator.

(c) (5 points) Use the generating functional to compute the four-point function

$$G_4(x_1, x_2, x_3, x_4).$$

Show explicitly that it is given by a sum over all pairings of two-point functions,

$$G_4 = G_F(x_1 - x_2)G_F(x_3 - x_4) + G_F(x_1 - x_3)G_F(x_2 - x_4) + G_F(x_1 - x_4)G_F(x_2 - x_3),$$

and briefly explain how this result follows from Wicks theorem for free fields.

Problem 2. ϕ^4 Theory, Diagrams, and LSZ (20 points)

Consider a real scalar field with interaction

$$\mathcal{L}_I = -\frac{\lambda}{4!}\phi^4.$$

- (a) (5 points) Draw all *connected* Feynman diagrams contributing to the two-point function $G_2(x_1, x_2)$ up to order λ^2 . Indicate clearly which diagrams are excluded due to vacuum bubble cancellation.
- (b) (5 points) For each contributing diagram in part (a), write down the corresponding *coordinate-space* expression, including all symmetry factors.
- (c) (10 points) Consider $2 \rightarrow 2$ scattering in this theory. Using the LSZ prescription:
- identify the leading-order contributing diagram,
 - explain why diagrams without interaction vertices do not contribute,
 - compute the scattering amplitude $M_{2 \rightarrow 2}$.

Your final answer should be expressed in terms of λ .